



Soumik Sahoo

B. Tech in Engineering Physics

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RESEARCH INTERESTS

My research interests lie in experimental quantum devices and quantum transport, particularly semiconductor spin qubits, charge sensing, cryogenic RF reflectometry, and coherent control. I am also interested in non-Hermitian and topological quantum systems, and in combining low-temperature measurements with analytical and numerical modeling for scalable quantum technologies.

Degree/Examination	University	Institute	Year	CPI/%
B. Tech in Engineering Physics	IIT Bombay	IIT Bombay	2023-2027	9.28/10
Higher Secondary +2	CHSE, Odisha	Saraswati Vidya Mandir, Neelakanthanagar	2022	89.66%
Secondary 10th	BSE, Odisha	Saraswati Vidya Mandir, Singhpur	2020	93.5%

- Pursuing an **Honors** degree in the **Department of Physics, IIT Bombay**
- Pursuing a **Minor** degree in **Centre for Machine Intelligence and Data Science, IIT Bombay**

SCHOLASTIC ACHIEVEMENTS

- Secured 98.2 percentile out of 0.2 million candidates in Joint Entrance Examination Advanced. (2023)
- Achieved 99.87 percentile out of 1.2 million candidates in Joint Entrance Examination Mains. (2023)

KEY PROJECTS

- **Low Temperature Qubit Characterization of Semiconductor Quantum Dots** [Link](#) (Aug '25 - Present)
Supervisor: Prof. Suddhasatta Mahapatra & Prof. Uditendu Mukhopadhyay | Department of Physics, IIT Bombay
 - Performed systematic tuning and electrostatic characterization of GaAs and Silicon quantum dot devices, optimizing gate voltage configurations to achieve stable few-electron regimes essential for qubit operation.
 - Implemented RF reflectometry measurement setups using Vector Network Analyzers and Lock-in Amplifiers for high-sensitivity dispersive readout of quantum dot charge states, enabling real-time charge transitions.
 - Developing pulse sequences on an Arbitrary Waveform Generator for 3-level pulsing protocols to probe spin relaxation time (T_1) in silicon quantum dots, contributing to benchmarking of qubit coherence properties.
 - Operated Bluefors dilution refrigerator for sample loading, wiring, and cooldown procedures, achieving stable millikelvin-regime measurements of GaAs and silicon quantum dot devices.
- **Induced Superconductivity in Magnetic Topological-Insulator Thin Films** [Draft](#) (Summer '26 - Present)
Research Internship: Prof. Thomas Schmidt | Department of Physics and Materials Science, University of Luxembourg | Manuscript in preparation
 - Developed a Green's-function theory for proximity-induced superconductivity in MTI thin films coupled to a conventional s -wave superconductor, deriving normal and anomalous propagators through fourth order in interface tunneling.
 - Proved the vanishing of odd perturbative orders through fermion-parity selection and reduced connected fourth-order Wick contractions into normal-scattering and Andreev-conversion topologies.
 - Evaluated superconducting kernels analytically through k_z contour integration and resummed all even tunneling orders using the Dyson equation to obtain a frequency-dependent 8×8 BdG effective Hamiltonian and pole equation.
 - Identified rank-one orbital-selective pairing in the symmetric-antisymmetric basis and implemented a four-momentum Pfaffian-sign diagnostic for Chern-parity changes in a 50-layer MTI slab.
- **Non-Hermitian Topological Systems** [Link](#) (Jan '26 - Present)
Supervisor: Prof. Lee Ching Hua | Department of Physics, National University of Singapore

- Investigated bosonic system caging governed by the 1D Non-Hermitian Skin Effect (NHSE), analyzing the macroscopic accumulation of localized eigenstates under open boundary conditions.
 - Expanding the theoretical framework into 2D lattice models, mathematically leveraging the algebraic skin effect to map anomalous boundary localization and multidimensional topological phenomena.
 - Constructed and diagonalized non-reciprocal tight-binding Hamiltonians to computationally verify complex energy spectra, phase transitions, and unique wave-packet dynamics.
- **Free Fermions in Disguise** [Report](#) (Aug '25 - Present)
Supervisor: Prof. Sumiran Pujari | Department of Physics, IIT Bombay
- Studied emergent free-fermion behavior in interacting spin chains using Fendley's Free Fermions in Disguise framework and derived mappings between spin and Majorana representations in 1D quantum models.
 - Investigated algebraic structures and symmetries underlying fermionization, connecting Clifford algebras and non-local operators to the emergence of solvable quantum phases.
 - Implemented numerical simulations and algebraic derivations to validate the free-fermion correspondence, visualizing energy spectra, degeneracies, and topological phases arising from non-trivial fermionization.
- **Electronic Transport and Scattering in Conductors** [Link](#) (Jan '25 - April'25)
Supervisor: Prof. Soumya Bera | Department of Physics, IIT Bombay
- Analyzed scattering in conductors by Boltzmann transport equation and relaxation time approximation.
 - Explored electron-phonon and electron-electron interactions, deriving temperature-dependent resistivity.
 - Formulated the momentum relaxation via impurity and Umklapp scattering processes.
- **Non-Equilibrium Green's Function Formalism** [Link](#) | Computational Physics Project (Dec '24 - Jan '25)
- Implemented NEGF formalism to model coherent electron transport in nanoscale systems, achieving accurate I-V predictions, capturing conductance quantization effects, and enhancing quantum device simulations.
 - Integrated Self-Consistent Field, bandstructure analysis, and subband modeling, visualizing key quantum phenomena such as density of states, local density of states, potential profiles, and transmission functions.

COURSE PROJECTS

- **Spectra of Kitaev Chains** [Presentation](#) (PH 554) (Aug'26-Present)
Prof. Soumya Bera | Department of Physics, IIT Bombay
- Formulated Kitaev-chain models in the many-body Fock basis and quadratic BdG Nambu basis, comparing Hilbert-space scaling, Hamiltonian structure, and low-energy eigenspectra for finite chains.
 - Computed Zeeman-dependent spectra, topological invariants, and finite-size scaling to distinguish true and false Poor Man's Majorana modes under homogeneous and site-dependent fields.
- **Chiral Tunnelling and the Klein Paradox in Graphene** [Presentation](#) (PH 436) (Aug'25-Nov'25)
Prof. Hridish Pal | Department of Physics, IIT Bombay
- Derived relativistic reflection and transmission amplitudes across electrostatic barriers, applying wave-vector matching and group-velocity corrections to resolve the Klein paradox at oblique incidence.
 - Constructed graphene's tight-binding and Bloch Hamiltonians near the Dirac points, linking pseudospin conservation and chirality to perfect normal-incidence Klein tunnelling and anti-Klein reflection in bilayers.
- **Orbital Hall Effect (PH 557)** [Presentation](#) (Aug'25-Nov'25)
Prof. Sayantika Bhowal | Department of Physics, IIT Bombay
- Compared quantum, spin, and orbital Hall effects through topology, symmetry constraints, characteristic band structures, and mechanisms generating transverse charge and angular-momentum currents.
 - Analyzed orbital-texture-driven transport and charge-orbital conversion, explaining how orbital currents emerge without net spin polarization and motivate efficient orbitronic and spintronic device concepts.
- **Orbital Dynamics of Similar and Dissimilar Objects** [Link](#) (PH 222) (Jan'25-April'25)
Prof. Pradeep Sarin | Department of Physics, IIT Bombay
- Built an Arduino Mega-TFT interface to simulate binary-star and satellite-planet orbital trajectories.
 - Validated numerical trajectories, energies, and periods against analytical results below 0.02% error.

- **Nonlinear Dynamics of the Chaotic Duffing Oscillator** [Report](#) (PH 567) (Aug'24-Dec'24)
Prof. Punit Paramananda | Department of Physics, IIT Bombay
 - Simulated driven Duffing dynamics across periodic, bifurcating, and chaotic regimes under varied forcing.
 - Linearized fixed points and visualized time series, phase portraits, Poincaré sections, and an analog circuit.
- **Physics-Informed Quantum System Analysis** [Report](#) (PH 227) (Aug'24-Dec'24)
Prof. Alok Shukla and Prof. Sadhana Dash | Department of Physics, IIT Bombay
 - Generated Schrödinger datasets and trained PINNs to predict quantum harmonic-oscillator wavefunctions.
 - Achieved MSE 0.00077, beating random forests (0.01295) and XGBoost (0.03213) on identical data.
- **Voice and Song Recognition** [Link](#) (DS 203) (Aug'24-Dec'24)
Prof. Vinay Kulkarni | CMINDS, IIT Bombay
 - Extracted MFCC features from 100+ vocal clips to identify singers, songs, and associated musical styles.
 - Benchmarked SVM, KNN, and random-forest classifiers for robust multiclass recognition across the dataset.

TECHNICAL SKILLS

- **Programming & Scripting Languages:** C++, Java, JavaScript, Python, HTML, CSS, Blender, Photoshop
- **Libraries:** QCoDeS, scikit-learn, Matplotlib, Pandas, NumPy, SciPy, Seaborn, Plotly, TensorFlow, Keras, XGBoost, PyTorch, MATLAB, $\LaTeX$, ReactJS, PHOEBE, LTspice, Figma
- **Tools:** Arbitrary waveform generator, dilution refrigerator, oscilloscope, vector network analyzer, breakout box

MENTORSHIP AND POSITION OF RESPONSIBILITY

- **Teaching Assistant In Introduction to Quantum Mechanics** (Jan'26 - Now)
 - I have hands-on experience teaching and solving tutorials for 30 undergraduates.
- **Teaching Assistant In Electronics Lab** (Aug'25 - Nov'25)
 - I have hands-on experience assisting over 60 Masters students in their foundational Electronics lab.
- **Introduction to Quantum Computing** (May'25 - July'25)
 - I have mentored 13 students during SOC on Quantum Informations and Computing for 6 weeks.

KEY COURSES UNDERTAKEN

- **Physics:** Analog and Digital Electronics, Quantum Mechanics I & II, Statistical Mechanics, Electromagnetic Theory, Quantum Information and Computing, Condensed Matter physics, Magnetism and Superconductivity, Computational Many Body Physics .
- **Mathematics:** Calculus, Linear Algebra, Differential Equation, Complex Analysis, and Integral transforms.
- **Miscellaneous:** Introduction to Psychology, Introduction to Management, Economics, Computer Utilization and Programming, AI and Data Science, Makerspace, Design thinking and Innovation.

EXTRACURRICULAR ACTIVITIES

Science Exhibition	• Secured top honors in district-level Vedic math competitions twice. (2020-21)
Social	• Devoted 80+ hours to NSS Green Campus by lots of planting activities in the campus. (2023-24)

REFERENCES

Prof. Suddhasatta Mahapatra
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Prof. Uditendu Mukhopadhyay
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